IAT New Test 3

Question paper, answer key
and worked solutions

> **MANUAL REVIEW DRAFT**
> 2027 preparation using the verified 2026 pattern and stored 2026 syllabus.
> The 2027 brochure has not been verified. Not approved for student publication.

60	180	240
single-correct questions	minutes	maximum marks

Subject	Question numbers	Marks
Biology	1-15	60
Chemistry	16-30	60
Mathematics	31-45	60
Physics	46-60	60

Instructions

1. There are 60 single-correct questions: Biology 1–15, Chemistry 16–30, Mathematics 31–45 and Physics 46–60.
2. Each question has exactly one correct option. Correct: +4; incorrect: -1; unanswered: 0. All questions are available to attempt; there is no best-three-subject scoring rule.
3. Use 180 minutes. An on-screen non-programmable scientific calculator is part of the 2026 examination conditions; personal calculators are not allowed under that scheme.
4. Treat stated data and idealisations as exact unless an approximation is explicitly supplied. Log denotes base 10 and ln denotes the natural logarithm.
5. The review PDF contains the question paper first, followed by a separate answer key and worked solutions. Keep the solutions hidden during a timed attempt.

Fixed authored order. Figures and IDs must remain linked during review.
Revised 2026-10-31 · IAT27_TEST_03 · Original AI-assisted content.

QUESTION PAPER / B

Biology

Questions 1-15 · 15 questions · 60 marks

Q1

+4 / -1

An alga has phycoerythrin, stores floridean starch and lacks motile reproductive cells. Which class is supported by all three observations?

A) Chlorophyceae

B) Phaeophyceae

C) Rhodophyceae

D) Bryophyta

Q2

+4 / -1

In a young root, radial xylem and phloem lie inside a cell layer whose radial walls contain a suberin-rich Casparian strip. What does that layer chiefly regulate?

A) Apoplastic entry into the stele by forcing selective membrane crossing

B) Photosynthetic gas exchange in mesophyll

C) Longitudinal phloem pressure from leaf to sink

D) Direct pollen transfer to a stigma

Q3

+4 / -1

A carrier imports Na^+ and glucose together even when cellular glucose is higher than outside. Blocking the ATP-driven Na^+ pump eventually stops glucose entry, although the carrier remains present. What powers uptake?

A) Glucose's own downhill gradient

B) Direct ATP hydrolysis at the glucose carrier is established by the result

C) Simple diffusion through membrane lipid

D) The Na^+ gradient maintained by the pump drives coupled glucose transport

Q4

+4 / -1

Identical illuminated leaves have equal stomatal aperture and leaf temperature. X is in humid air and Y in dry air, with equal air movement. Initially, which loses more water vapour and why?

A) X, because humid air pulls vapour more strongly

B) Y, because its outward vapour-pressure difference is larger

C) Both equally, because aperture alone sets diffusion

D) Neither, because transpiration requires darkness

Q5

+4 / -1

Two tissues receive blood of the same arterial oxygen content. Tissue Q produces more CO_2 and has lower local pH than P. What haemoglobin response helps deliver oxygen to Q?

A) Higher pH at Q prevents release

B) Haemoglobin loses all binding capacity permanently

C) Lower pH favours unloading at a given oxygen partial pressure

D) CO_2 creates additional oxygen in blood

Q6

+4 / -1

Bagged flowers of P set no seed after hand self-pollination, but set seed after compatible cross-pollination. P's pollen works on another compatible plant. What inference is supported by these controls?

A) Failure is specific to the self combination, consistent with self-incompatibility

B) All P pollen is universally nonviable

C) P ovules never develop following fertilisation

D) Bagging always prevents seed formation

Q7

+4 / -1

The figure shows a short DNA duplex and a promoter arrow pointing right to left. Assume transcription starts at the rightmost shown base, uses the strand read 3′→5′, and ends after the shown sequence. Which mRNA is made, written 5′→3′?

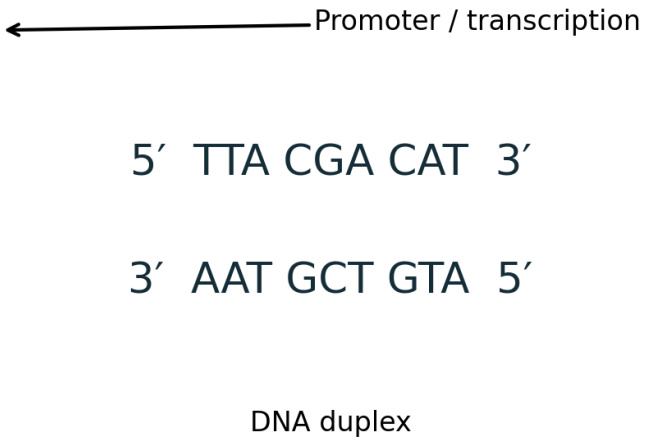


Figure for Q7

A) 5′-UUA CGA CAU-3′

B) 5′-AAU GCU GUA-3′

C) 5′-UAC AGC AUU-3′

D) 5′-AUG UCG UAA-3′

Q8

+4 / -1

A person loses many helper T lymphocytes while neutrophil numbers initially remain normal. Which impairment best follows from the helper-cell role?

A) Only clot formation is lost

B) Coordination of both antibody and cell-mediated adaptive responses weakens

C) All pathogens immediately stop dividing

D) Innate neutrophils become the sole source of antigen-specific antibodies

---

Q9

+4 / -1

PCR tubes X and Y have the same template, polymerase, nucleotides and cycling. X has both primers flanking a target; Y lacks the reverse primer. What follows for the defined double-stranded target after many ideal cycles?

A) Y must amplify the bounded target more rapidly

B) Both amplify exponentially at the same rate

C) X can amplify it exponentially; Y cannot sustain that bounded two-primer process

D) Neither can extend any DNA strand

Q10

+4 / -1

A plant community has 1000 units net production in an interval. Herbivores consume 250 units and 100 units remain as unconsumed plant biomass; all the rest enters detritus. How many units enter detritus?

A) 650

B) 750

C) 900

D) 1150

Q11

+4 / -1

Pigment needs two independent functional enzymes encoded by A and B; aa or bb blocks the pathway. Two AaBb plants cross. What fraction of offspring is pigmented under complete dominance?

A) 3/4

B) 1/4

C) 7/16

D) 9/16

Q12

+4 / -1

In strain P, a functional repressor cannot bind its mutated operator; promoter and structural genes work. Strain Q has a normal operator and repressor. With no glucose, compare enzyme expression without and with lactose.

A) Both express only without lactose

B) P expresses in both; Q is low without lactose and induced with it

C) P is always off; Q is always on

D) Both remain off because a promoter is present

Q13

+4 / -1

Presynaptic stimulation releases normal transmitter in two preparations. A postsynaptic response occurs in P but not Q; direct electrical stimulation of Q's postsynaptic membrane still gives a normal action potential. Where is the likely defect?

A) Presynaptic transmitter synthesis

B) Axonal action-potential generation in Q's postsynaptic membrane

C) Postsynaptic transmitter reception or coupling

D) Complete absence of the presynaptic neuron

Q14

+4 / -1

In ordinary meiosis, when does chromosome DNA replication occur relative to meiosis I and II?

A) Once before meiosis I, without replication between the divisions

B) Once before each division

C) Only after meiosis II

D) Never, because chromosomes are already visible

Q15

+4 / -1

The figure shows one inherited marker in mother M, child C and candidates P and Q. Assume one allele from each biological parent, precise band sizing and no mutation. What follows from this marker alone?

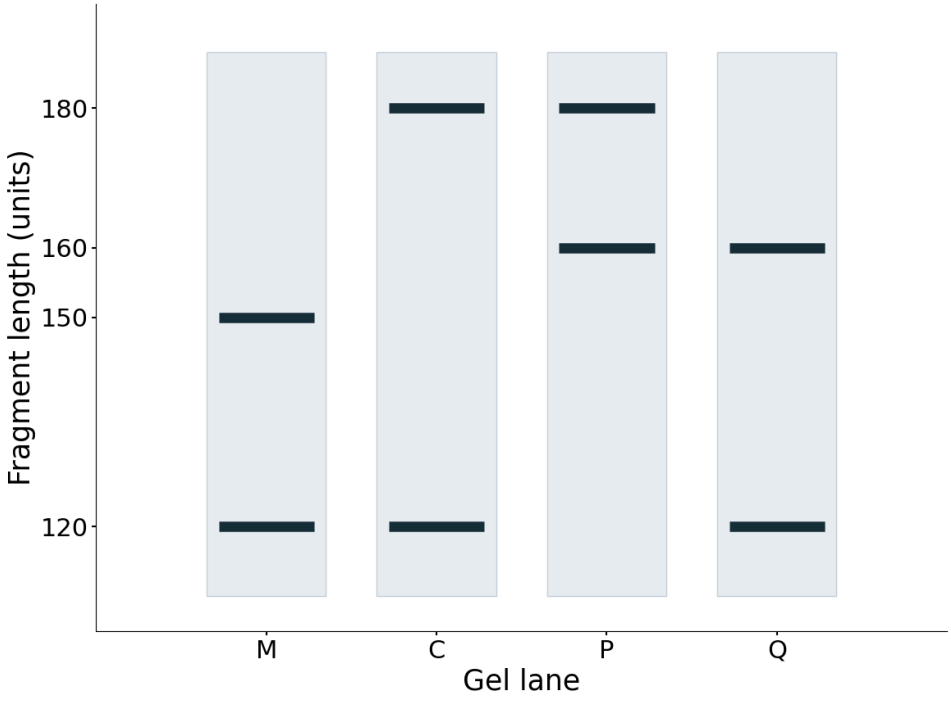


Figure for Q15

A) Q is established as father with certainty

B) P is compatible and Q excluded, but P is not uniquely proved by one marker

C) Both are excluded because C shares a band with M

D) Each candidate is certainly a parent because each has two bands

Chemistry

Questions 16-30 · 15 questions · 60 marks

Q16

+4 / -1

Each C–O bond in CO_2 is polar. Under its linear geometry, what is the molecule's net dipole moment?

A) Equal to one bond dipole

B) Zero because equal opposite bond dipoles cancel

C) Twice one bond dipole along the same direction

D) Undefined because carbon is central

Q17

+4 / -1

The figure shows small octahedral splitting for complex X and large splitting for Y, both d^6 at the same metal oxidation state. Under the simple crystal-field model, how many unpaired electrons do X and Y have?

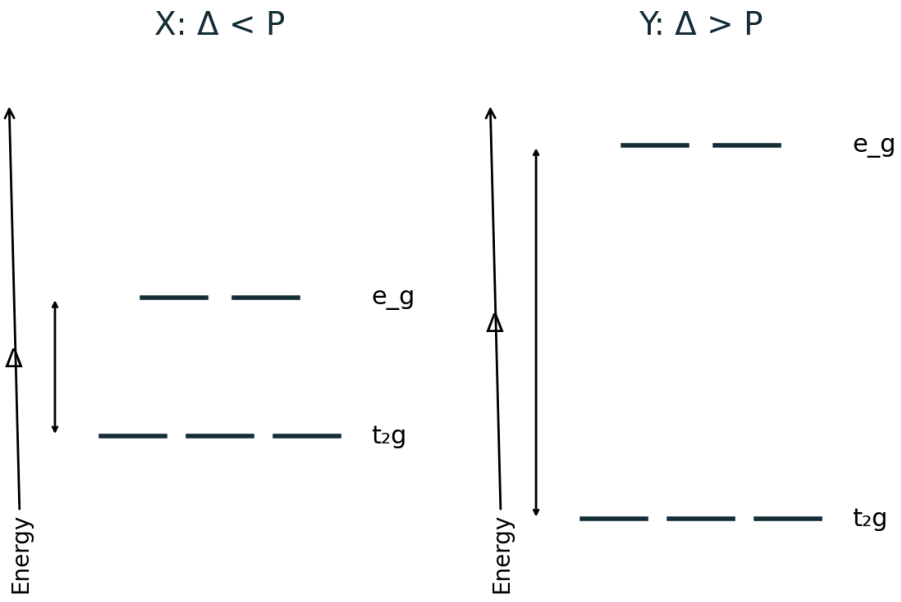


Figure for Q17

A) 0 and 4

B) 2 and 2

C) 6 and 0

D) 4 and 0

Q18

+4 / -1

Between Mn^{2+} (Z=25) and Fe^{2+} (Z=26), which has a half-filled 3d subshell and how many 3d electrons does it contain?

A) Mn^{2+}, five

B) Fe^{2+}, five

C) Mn^{2+}, seven

D) Fe^{2+}, six

Q19

+4 / -1

Why is first ionisation energy of N higher than O, despite O's larger nuclear charge?

A) Nitrogen has a complete $2p^6$ subshell

B) Oxygen has no 2p electrons

C) Oxygen has a paired 2p orbital with extra repulsion relative to N's $2p^3$ arrangement

D) Nitrogen's outer electrons occupy n=3

Q20

+4 / -1

A proton of mass m, charge e and alpha particle of mass 4m, charge 2e start from rest and traverse the same accelerating potential V nonrelativistically. Find $\lambda_{proton}/\lambda_{\alpha}$.

A) 2

B) $\sqrt{8}$

C) $\sqrt{2}$

D) 8

Q21

+4 / -1

A+B→C uses 0.10 mol A and 0.16 mol B in a 1:1 reaction, isolating C at 80% theoretical yield. Then 2C→D isolates D at 75% of the theoretical yield from the isolated C. How many moles D result?

A) 0.060

B) 0.040

C) 0.075

D) 0.030

Q22

+4 / -1

At the para position, compare nitro- and methoxy-substituted benzoic acids with benzoic acid under the same conditions. Which is expected to be strongest?

A) p-Nitrobenzoic acid, because the electron-withdrawing group stabilises benzoate

B) p-Methoxybenzoic acid, because resonance donation stabilises benzoate

C) Benzoic acid, because substitution can never affect acidity

D) All exactly equal, because each has a COOH group

Q23

+4 / -1

The figure shows the two distinct carbonyl products from reductive ozonolysis of one acyclic alkene molecule, one molecule of each. Which alkene reconstructs the original carbon skeleton?

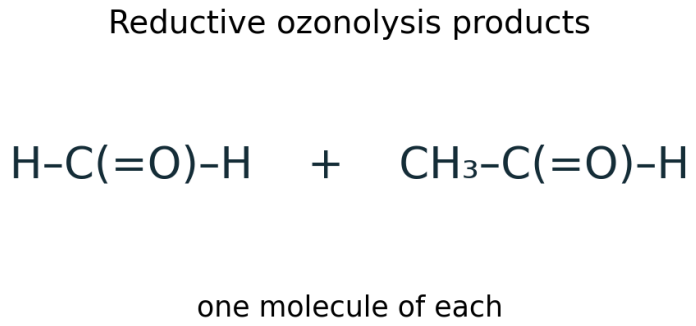


Figure for Q23

A) Ethene

B) But-2-ene

C) Propene

D) 2-methylpropene

Q24

+4 / -1

Under mild conditions that permit ordinary SN2 substitution of a primary bromoalkane by hydroxide, bromobenzene is much less reactive. Which structural distinction matters most?

A) Bromobenzene has no C–Br bond

B) Resonance gives aryl C–Br partial double-bond character and sp^2 carbon disfavors backside SN2 attack

C) A primary bromoalkane has a benzene ring

D) Bromobenzene is freely ionised into phenyl cations in water

Q25

+4 / -1

Phenol forms a salt with ordinary aqueous NaOH, whereas ethanol is not comparably deprotonated. What explains the difference?

A) Ethanol contains no O–H bond

B) Phenol is an aldehyde

C) Hydroxide reacts only with aromatic C–C bonds

D) Phenoxide is resonance-stabilised relative to an alkoxide

Q26

+4 / -1

Complete hydrolysis of Gly–Ala yields free glycine and alanine. At pH well above both isoelectric points, what net-charge tendency does each free amino acid have?

A) Both tend net negative

B) Both tend net positive

C) Glycine positive, alanine negative solely due to side chains

D) Both must be net neutral at every pH

Q27

+4 / -1

AgCl has $K_{sp}=1.0\times10^{-10}$ at the stated temperature. Approximate equilibrium $[Ag^+]$ in 0.10 M NaCl if its dissolution changes $[Cl^-]$ negligibly.

A) 1.0×10^{-5} M

B) 1.0×10^{-10} M

C) 1.0×10^{-9} M

D) 0.10 M

Q28

+4 / -1

At fixed temperature and composition, a liquid mixture's total vapour pressure exceeds its ideal Raoult-law prediction. Which interaction pattern is consistent with this positive deviation?

A) Unlike A–B attractions must be stronger than like pairs

B) Unlike A–B attractions are weaker on average, permitting easier escape

C) The mixture has zero vapour pressure

D) All molecular attractions vanish

Q29

+4 / -1

Aqueous $CuSO_4$ is electrolysed with inert Pt electrodes under conditions where Cu^{2+} deposits at the cathode and water is oxidised at the anode. Which primary products follow?

A) H_2 and Cu

B) Cu and SO_2

C) Cu and gaseous sulfate

D) Cu and O_2

Q30

+4 / -1

50.0 mL of 1.00 M monoprotic acid and 50.0 mL of 0.500 M strong base react fully in an insulated calorimeter. The final 100 g solution warms 6.0 °C; use c=4.20 J g^{-1} K^{-1} and neglect calorimeter heat. Assuming only neutralisation causes the measured heat, find enthalpy per mole water formed.

A) −100.8 kJ/mol

B) −50.4 kJ/mol

C) +100.8 kJ/mol

D) −201.6 kJ/mol

QUESTION PAPER / M

Mathematics

Questions 31-45 · 15 questions · 60 marks

Q31

+4 / -1

The circle $x^2+y^2-6x-8y=0$ is touched by a line $3x+4y=k$. What are the possible values of k?

A) -25 and 25

B) -5 and 45

C) 25 only

D) 0 and 50

Q32

+4 / -1

The point (1,1) lies on $x^2+xy+y^2=3$. Where does its tangent meet the x-axis?

A) $x=-2$

B) $x=1$

C) $x=2$

D) It is parallel to the x-axis

Q33

+4 / -1

Equal squares of side x are cut from the four corners of a 12 cm by 8 cm rectangle, and the sides are folded into an open box. Which x gives maximum volume?

A) $(10+2\sqrt{7})/3$ cm

B) $(10-2\sqrt{7})/3$ cm

C) 2 cm

D) 4 cm

Q34

+4 / -1

Evaluate $\int_0^1[x/(1+x^2)-x/(1+x)]dx$.

A) $(3/2)\ln 2-1$

B) $1-(3/2)\ln 2$

C) $(1/2)\ln 2$

D) $\ln 2-1$

Q35

+4 / -1

Let $f(x)=\sqrt{(x-1)}$ and $g(t)=1/(t-2)$, both real-valued where defined. What is the domain of $g \circ f$?

A) $x>1$ only

B) $x \geq 5$ only

C) All real x except 5

D) $x \geq 1$ with $x \neq 5$

Q36

+4 / -1

Vectors $a=(1,1,0)$, $b=(1,0,1)$, $c=(t,1,1)$ span a parallelepiped of volume 3. What real t-values are possible?

A) 2 only

B) -5 and 1

C) -1 and 5

D) 1 and 5

Q37

+4 / -1

A has rows R_1, R_2, R_3 and det A=3. Matrix B has rows R_1+R_2, R_2+R_3, R_3+R_1 in that order; then its first and third rows are interchanged. Find det B.

A) +6

B) -6

C) -3

D) +3

Q38

+4 / -1

Three balls are drawn together from a bag with 4 red and 3 blue balls. Given that at least two drawn balls are red, what is the probability that all three are red?

A) 2/11

B) 4/35

C) 1/2

D) 3/11

Q39

+4 / -1

A path from (0,0) to (5,4) uses only unit right or up steps. How many such paths avoid the point (2,2)?

A) 126

B) 60

C) 36

D) 66

Q40

+4 / -1

For integer $n \geq 4$, the coefficient of x^3 in $(1+x)^n(1-x)$ vanishes. What is the coefficient of x^4 in the same polynomial?

A) +5

B) −10

C) −5

D) 0

Q41

+4 / -1

What is $\Sigma_{k=1}^{4}\ 1/[k(k+2)]$?

A) 11/24

B) 17/30

C) 21/40

D) 4/5

Q42

+4 / -1

For which real k does $x^2-(k+1)x+(k-2)=0$ have one negative root and one positive root strictly greater than 2?

A) $0<k<2$

B) $k<2$

C) $0 \leq k \leq 2$

D) $k>2$

Q43

+4 / -1

Two equally sized groups have means 1 and 5 and population variances 4 and 4. What is the population variance when all observations are pooled?

A) 4

B) 6

C) 12

D) 8

Q44

+4 / -1

Which differential equation is satisfied by every function $y=C_1e^x+C_2e^{(-x)}$, for arbitrary constants C_1, C_2?

A) $y'=y$

B) $y''=-y$

C) $y''=y$

D) $y'=0$

Q45

+4 / -1

Define $f(x)=\sin(2x)/x$ for $x\neq 0$ and $f(0)=k$. For continuity at zero, what are k and $f'(0)$?

A) $k=0, f'(0)=2$

B) $k=2, f'(0)=0$

C) $k=2, f'(0)=2$

D) No value of k works

Physics

Questions 46-60 · 15 questions · 60 marks

Q46

+4 / -1

From the same height, one ball is released horizontally and another is dropped from rest at the same instant. Neglect air resistance. Which lands first on level ground?

A) Both simultaneously

B) Horizontally released ball

C) Dropped ball

D) It depends on horizontal speed

Q47

+4 / -1

At an instant, $v=(3\hat{i}+4\hat{j})$ m/s and force $F=(2\hat{i}-\hat{j})$ N. What instantaneous power does this force supply?

A) 10 W

B) 5 W

C) −2 W

D) 2 W

Q48

+4 / -1

The figure shows a uniform disc of original mass M and radius R with a circular hole of radius R/2 removed; its centre is R/2 from original centre O. About the perpendicular axis through O, what is the remaining moment of inertia?

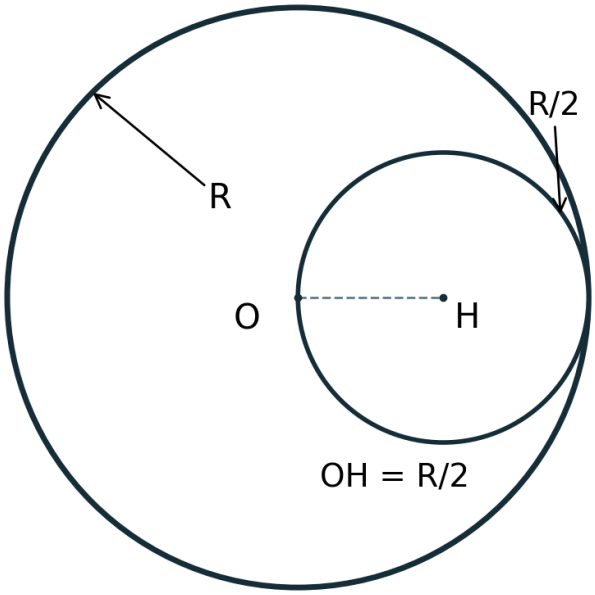


Figure for Q48

A) 3MR²/8

B) 15MR²/32

C) 13MR²/32

D) 7MR²/16

Q49

+4 / -1

An open water tank accelerates to the right at g/2. In its frame, what is the magnitude of the steady free-surface slope, vertical rise divided by horizontal distance?

A) 2

B) 1/2

C) 1

D) 1/4

Q50

+4 / -1

Two identical bodies of constant heat capacity C start at absolute temperatures T and 3T. Isolated together, they equilibrate without work. What is their total entropy change?

A) C ln(4/3)

B) 0

C) C ln(3/4)

D) 2C ln2

Q51

+4 / -1

At a fixed instant, two points along a sinusoidal travelling wave are separated by λ/6 in the propagation direction. What is their phase difference modulo 2π?

A) π/6

B) π/2

C) 2π/3

D) π/3

Q52

+4 / -1

A satellite of mass m orbits a planet of mass M in a circle of radius r. Taking potential energy zero at infinity, what is its total mechanical energy?

A) +GMm/(2r)

B) −GMm/r

C) −GMm/(2r)

D) 0

Q53

+4 / -1

The figure places +q at x=0 and −4q at x=3a, with a>0. At which finite points on the x-axis is the electric potential zero?



Figure for Q53

A) x=a and x=−3a

B) x=−a and x=3a/5

C) x=0 and x=3a

D) x=−3a/5 and x=a

Q54

+4 / -1

A parallel-plate capacitor stays connected to an ideal constant-voltage battery while plate separation is doubled, with area and dielectric unchanged. How do stored charge and energy change?

A) Both become half

B) Both double

C) Charge halves but energy doubles

D) Charge stays fixed and energy halves

Q55

+4 / -1

A cell of emf 12 V and internal resistance 2 Ω supplies a 4 Ω external resistor. What is the terminal voltage while it delivers current?

A) 12 V

B) 4 V

C) 6 V

D) 8 V

Q56

+4 / -1

The plane of a single-turn loop of area A carrying current I makes 30° with a uniform magnetic field B. What is the torque magnitude?

A) IAB/2

B) IAB

C) (√3/2)IAB

D) Zero

Q57

+4 / -1

An ideal transformer has 200 primary turns and 50 secondary turns. Its primary is at 240 V rms; the secondary drives a 6 Ω resistive load. What are secondary voltage and primary current?

A) 60 V and 10 A

B) 60 V and 2.5 A

C) 240 V and 2.5 A

D) 120 V and 5 A

Q58

+4 / -1

A ray crosses a slab of thickness L normally, refractive index n. Relative to the same geometric distance in vacuum, what is excess optical path length?

A) (n−1)L

B) nL

C) L/n

D) (n+1)L

Q59

+4 / -1

An electromagnetic plane wave has electric field along +y and magnetic field along +z. What direction does it propagate?

A) −x

B) +y

C) +z

D) +x

Q60

+4 / -1

Equal initial numbers N_0 of nuclides X and Y have half-lives 2 h and 4 h. After 8 h, find (i) activity ratio AX:AY and (ii) fraction of all initial nuclei still undecayed.

A) 1:4 and 5/32

B) 2:1 and 5/32

C) 1:2 and 5/32

D) 1:2 and 27/32

Answer key

One correct option per question. Question and option order are fixed.

Biology

Q	Answer
1	**C**
2	**A**
3	**D**
4	**B**
5	**C**
6	**A**
7	**D**
8	**B**
9	**C**
10	**A**
11	**D**
12	**B**
13	**C**
14	**A**
15	**B**

Chemistry

Q	Answer
16	**B**
17	**D**
18	**A**
19	**C**
20	**B**
21	**D**
22	**A**
23	**C**
24	**B**
25	**D**
26	**A**
27	**C**
28	**B**
29	**D**
30	**A**

Mathematics

Q	Answer
31	**D**
32	**C**
33	**B**
34	**A**
35	**D**
36	**C**
37	**B**
38	**A**
39	**D**
40	**C**
41	**B**
42	**A**
43	**D**
44	**C**
45	**B**

Physics

Q	Answer
46	**A**
47	**D**
48	**C**
49	**B**
50	**A**
51	**D**
52	**C**
53	**B**
54	**A**
55	**D**
56	**C**
57	**B**
58	**A**
59	**D**
60	**C**

Reviewer checklist

Accuracy and single correct answer · Syllabus fit · Distractor quality
Diagram labels · Solution clarity · Difficulty and timed attempt

Reviewer: _____ Date: _____

Decision / notes: _____

Editorial difficulty mix: 9 easy / 42 medium / 9 hard. These labels require human review and are not official item statistics.

Biology

Method, reasoning and common errors

Q1

Biology · easy

Diversity In the Living World / Algal pigments and reserve food

Answer: (C) Rhodophyceae.

Why: Red algae commonly possess phycoerythrin and store floridean starch. The lack of flagellated reproductive stages supplies a third consistent clue, so classification does not rest on visible colour alone.

Why not the others: (A) Green algae characteristically store starch and chlorophyll predominates. (B) Brown algae have fucoxanthin and laminarin/mannitol reserves. (D) Bryophytes are embryophyte land plants, not this algal class.

Key point: Combine pigments, storage material and life-cycle characters.

Q2

Biology · medium

Structural Organisation In Plants And Animals / Root endodermis and Casparian strip

Answer: (A) Endodermal selective entry.

Why: Water moving through cell walls encounters the waterproof strip at the endodermis. To pass inward toward the stele, it must cross a living membrane, where transport can be regulated. The radial vascular arrangement and wall band identify the location.

Why not the others: (B) Mesophyll is in leaves. (C) Phloem transport occurs after vascular entry and is not the strip's role. (D) Is reproductive transport unrelated to roots.

Key point: Relate wall modification to the path water must take.

Q3

Biology · medium

Cell: Structure And Functions / Secondary active transport

Answer: (D) Coupling to the Na^+ gradient.

Why: Glucose travels uphill, so passive diffusion cannot explain it. The pump uses ATP to maintain a downhill Na^+ gradient; cotransport can harness that gradient. Blocking the pump removes the stored gradient over time even while the cotransporter protein remains.

Why not the others: (A) Glucose is stipulated uphill. (B) The data do not show ATP acting directly on this carrier. (C) Lipid diffusion neither couples Na^+ nor sustains uphill glucose movement.

Key point: Distinguish direct ATP use from gradient-mediated secondary transport.

Q4

Biology · medium

Plant Physiology / Vapour-pressure gradient and transpiration

Answer: (B) Y in dry air.

Why: Water vapour is near saturation in internal leaf spaces. Lower external humidity creates a larger concentration or vapour-pressure gradient through otherwise similar stomata, increasing outward diffusion initially. Holding temperature, aperture and movement constant isolates humidity.

Why not the others: (A) Humidity reduces the gradient. (C) Equal aperture does not eliminate a changed driving difference. (D) Light is present and darkness is not required.

Key point: Flux depends on both conductance and driving gradient.

Q5

Biology · medium

Human Physiology / Bohr effect in active tissue

Answer: (C) Enhanced oxygen unloading.

Why: More CO_2 and acidity shift haemoglobin's oxygen dissociation relation toward lower affinity. At the same local oxygen pressure, more bound oxygen can therefore be released to metabolically active Q. This is a reversible local response, not destruction of haemoglobin.

Why not the others: (A) Reverses the stated pH change. (B) Falsely makes modulation permanent. (D) CO_2 does not synthesize oxygen.

Key point: Relate metabolic CO_2 production to oxygen delivery.

Q6

Biology · medium

Reproduction / Self-incompatibility controls

Answer: (A) Self-combination rejection.

Why: Cross pollen produces seed on P, establishing that its ovules and bagging procedure can support reproduction. P pollen succeeds elsewhere, establishing viability. The remaining contrast is the self combination, consistent with a self-incompatibility response.

Why not the others: (B) Contradicts successful P pollen use. (C) Contradicts cross seed set. (D) Bagged hand crossing worked.

Key point: Reciprocal controls isolate partner specificity from general infertility.

Q7

Biology · hard

Genetics And Evolution / Transcription direction and antiparallel DNA strands

Answer: (D) 5′-AUG UCG UAA-3′.

Why: Moving right to left, polymerase must read a 3′→5′ template, which is the top strand in this orientation. Reverse its base order to TAC AGC ATT, then make the complementary RNA AUG UCG UAA. Equivalently, read the bottom coding strand right to left and replace T by U. Both direction and complementarity matter.

Why not the others: (A) Copies the displayed top strand in the wrong direction. (B) Copies the bottom strand left to right. (C) reverses without making the RNA complement.

Key point: Select template by promoter direction before transcribing antiparallel bases.

Q8

Biology · medium

Biology In Human Welfare / Helper T-cell coordination

Answer: (B) Broad adaptive coordination weakens.

Why: Helper T cells signal to other immune cells, including B cells and cytotoxic T cells. Their sustained loss can thus impair both antibody production and cell-mediated responses even when a neutrophil count is initially normal. A count of one innate cell type does not replace adaptive coordination.

Why not the others: (A) Clotting is not the central helper-cell function. (C) No immune loss sterilises every pathogen. (D) Neutrophils do not secrete antigen-specific antibody.

Key point: Trace effects through the cells a helper population coordinates.

Q9

Biology · medium

Biotechnology / Two-primer PCR amplification

Answer: (C) Exponential amplification requires both primers.

Why: Opposed primers define both ends and allow newly synthesized target strands to serve as templates on subsequent cycles. With one missing primer, extension from the remaining primer may occur, but the same bounded double-stranded product cannot repeatedly generate itself by two-primer cycling.

Why not the others: (A) Missing a primer does not improve target amplification. (B) Overlooks the second defined end. (D) One primer can still support some extension.

Key point: Distinguish any extension from exponential amplification of a defined amplicon.

Q10

Biology · medium

Ecology / Energy partition into detritus

Answer: (A) 650 units.

Given: The prompt partitions net production into consumption, retained biomass and detritus.

Formula/Principle: Conservation under the stated exhaustive partition.

Step 1: Accounted elsewhere=250+100=350 units.

Step 2: Detritus=1000−350=650 units.

Why: Detritus is not identical to every unconsumed unit because some plant material remains stored during the interval.

Why not the others: (B) ignores stored biomass; (C) ignores herbivory; (D) adds a flow to net production.

Key point: Check whether categories overlap before subtracting.

Q11

Biology · medium

Genetics And Evolution / Complementary gene action

Answer: (D) 9/16.

Given: Pigment requires A_ and B_ together.

Formula/Principle: Independent segregation allows multiplication of the two dominant-phenotype probabilities.

Step 1: Each Aa×Aa locus gives functional phenotype probability 3/4.

Step 2: Both present=$(3/4)^2$=9/16.

Why: Either homozygous recessive block prevents the final pigment even if the other enzyme is present.

Why not the others: (A) checks only one locus; (B) treats two recessive classes as the requirement; (C) is the white fraction.

Key point: Follow the whole sequential pathway, not one gene alone.

Q12

Biology · hard

Genetics And Evolution / Lac operon operator defect

Answer: (B) P constitutive, Q inducible.

Why: In P the repressor cannot occupy its DNA site even without inducer, so transcription is not blocked by this control. In Q, bound repressor suppresses transcription until lactose-derived inducer reduces its binding. With glucose absent, the comparison isolates the operator–repressor response rather than catabolite effects.

Why not the others: (A) Reverses induction. (C) Reverses both strain behaviours. (D) A working promoter permits transcription rather than forbidding it.

Key point: Separate a defective DNA binding site from a defective protein.

Q13

Biology · medium

Human Physiology / Synaptic receptor versus membrane excitability

Answer: (C) Postsynaptic reception or coupling.

Why: Normal measured release argues against presynaptic failure. Direct electrical stimulation shows that the postsynaptic membrane can still generate an action potential. The missing response lies between arrival of transmitter and electrical activation, such as receptor binding or its coupling to channels.

Why not the others: (A) and (D) contradict normal transmitter release. (B) contradicts successful direct electrical activation.

Key point: Use a bypass experiment to localise a failure in a signalling chain.

Q14

Biology · easy

Cell: Structure And Functions / Replication before the two meiotic divisions

Answer: (A) Replication precedes meiosis I only.

Why: S phase copies each chromosome into sister chromatids before homologues separate in the first division. Meiosis II separates those sisters without a second intervening DNA-replication round, allowing haploid products.

Why not the others: (B) Would duplicate DNA again between divisions. (C) Comes too late for segregation. (D) Visible chromosomes have already been replicated.

Key point: Distinguish one replication from two consecutive divisions.

Q15

Biology · medium

Biotechnology / DNA marker parentage exclusion

Answer: (B) P compatible, Q excluded.

Why: The child can inherit 120 from M; the 180 allele must then come from the other biological parent. P has 180 and Q does not, so Q is excluded under the assumptions. Many untested people might carry 180, so a single marker cannot establish P uniquely.

Why not the others: (A) Q lacks 180. (C) Sharing a maternal allele is expected. (D) Two bands alone show neither transmission nor identity.

Key point: A marker may exclude a candidate without proving another uniquely.

Chemistry

Method, reasoning and common errors

Q16

Chemistry · easy

Chemical Bonding and Molecular Structure / Bond dipoles and molecular geometry

Answer: (B) Zero.

Why: The two identical bond dipole vectors lie on the same line but point in opposite directions from the central carbon. Vector addition cancels them even though each individual C–O bond remains polar. Molecular polarity depends on geometry as well as bond polarity.

Why not the others: (A) Keeps only one bond. (C) Adds magnitudes without directions. (D) A central atom does not prevent vector addition.

Key point: Sum bond dipoles as vectors.

Q17

Chemistry · hard

Coordination Compounds / High- and low-spin octahedral d6

Answer: (D) Four and zero.

Why: The small splitting in X makes occupation of the upper eg levels preferable to pairing too early: high-spin d^6 is $t_2g^4e_g^2$, with four unpaired electrons. The large gap in Y favours pairing in t_2g first: $t_2g^6e_g^0$ has no unpaired electrons. The figure supplies the energetic comparison but leaves the filling to be reasoned out.

Why not the others: (A) Reverses complexes. (B) Counts only eg electrons in X and invents unpaired electrons in Y. (C) Treats all six d electrons as independently unpaired despite only five orbitals.

Key point: Compare Δ with pairing energy before filling orbitals.

Q18

Chemistry · medium

The d- and f-Block Elements / Half-filled d subshell in common ions

Answer: (A) Mn^{2+}, $3d^5$.

Why: Neutral Mn is $[Ar]3d^5 4s^2$ and neutral Fe is $[Ar]3d^6 4s^2$. Forming 2+ ions removes the two 4s electrons first, leaving Mn^{2+} $3d^5$ and Fe^{2+} $3d^6$. Five d electrons give a half-filled five-orbital subshell.

Why not the others: (B) Assigns Mn's count to Fe. (C) Retains 4s electrons as 3d. (D) Correctly counts Fe but six is not half-filled.

Key point: Remove ns electrons before (n−1)d in these cations.

Q19

Chemistry · medium

Classification of Elements and Periodicity / Nitrogen–oxygen ionisation anomaly

Answer: (C) Pairing repulsion in O.

Why: N has one electron in each of the three 2p orbitals, a relatively stable half-filled arrangement. O adds a fourth 2p electron that pairs in one orbital; repulsion makes removal of one electron easier than a simple nuclear-charge trend suggests.

Why not the others: (A) N is $2p^3$, not $2p^6$. (B) O is $2p^4$. (D) Both valence shells are n=2.

Key point: Consider orbital occupancy when a periodic trend has an exception.

Q20

Chemistry · medium

Structure of Atom / de Broglie wavelengths after acceleration

Answer: (B) √8.

Given: Kinetic energy of each is qV; alpha mass and charge factors are 4 and 2.

Formula/Principle: $\lambda = h/\sqrt{(2mqV)}$.

Step 1: $p\alpha^2/pp^2 = (4m \times 2e)/(m \times e) = 8$.

Step 2: Since λ is inverse momentum, $\lambda p/\lambda\alpha = \sqrt{8}$.

Why: Equal potential does not mean equal kinetic energy for unequal charges.

Why not the others: (A) takes only mass ratio's square root; (C) takes only charge ratio; (D) omits the square root.

Key point: Include both mass and charge in accelerated-particle momentum.

Q21

Chemistry · medium

Some Basic Concepts of Chemistry / Limiting reagent with consecutive yields

Answer: (D) 0.030 mol.

Given: A limits the first reaction; the two yields apply successively.

Formula/Principle: Use stoichiometric ratios before applying each fractional yield.

Step 1: Theoretical C=0.10 mol, isolated $C = 0.10 \times 0.80 = 0.080$ mol.

Step 2: Theoretical $D = 0.080/2 = 0.040$ mol, isolated $D = 0.040 \times 0.75 = 0.030$ mol.

Why: The second stage can use only C actually isolated, not the first stage's hypothetical maximum.

Why not the others: (A) misses the 2:1 step; (B) omits second yield; (C) applies yield without the reaction ratio.

Key point: Track material through each stage separately.

Q22

Chemistry · medium

Organic Chemistry – Some Basic Principles and Techniques / Substituent effects on benzoic-acid strength

Answer: (A) p-Nitrobenzoic acid.

Why: Nitro withdraws electron density through inductive and resonance effects and helps stabilise the negatively charged conjugate base. At para, methoxy resonance donation tends to oppose this stabilisation. Acid strength follows how favourably deprotonation products are stabilised under comparable conditions.

Why not the others: (B) Donation does not stabilise the anion in the claimed way. (C) Ring substituents transmit electronic effects. (D) The same acidic group can have different environments.

Key point: Compare conjugate-base stabilisation, with substituent position specified.

Q23

Chemistry · medium

Hydrocarbons / Alkene reconstruction from ozonolysis fragments

Answer: (C) Propene.

Why: Each carbonyl carbon was originally one carbon of the C=C bond. Rejoining the methanal carbon and ethanal carbon by a double bond, replacing their carbonyl oxygens, gives $CH_2{=}CH{-}CH_3$. The fragments account for all three carbons in propene.

Why not the others: (A) Gives two methanals. (B) Gives two ethanals. (D) Gives methanal plus propanone rather than ethanal.

Key point: Reconstruct the two former alkene carbons from their separate carbonyl products.

Q24

Chemistry · medium

Haloalkanes and Haloarenes / Aryl versus alkyl substitution

Answer: (B) Aryl C–Br is resistant to ordinary SN2.

Why: Conjugation with the ring gives the C–Br bond partial double-bond character, and backside approach to an sp^2 aryl carbon is geometrically unfavourable. These features distinguish aryl from primary alkyl halides under the stipulated mild conditions.

Why not the others: (A) The bond exists. (C) Primary bromoalkane is not aromatic. (D) Stable free phenyl cations do not form readily in water.

Key point: Do not transfer an alkyl-halide mechanism uncritically to an aryl halide.

Q25

Chemistry · easy

Alcohols, Phenols and Ethers / Phenol acidity relative to ethanol

Answer: (D) Phenoxide stabilisation.

Why: Deprotonating phenol gives an anion whose charge can delocalise into the aromatic system. Ethoxide lacks that resonance stabilisation, so ethanol is a substantially weaker acid under the usual qualitative comparison.

Why not the others: (A) Ethanol has an O–H group. (B) Phenol is not an aldehyde. (C) The relevant reaction is proton transfer at oxygen.

Key point: Compare conjugate bases to explain an acidity contrast.

Q26

Chemistry · medium

Biomolecules / Charge of hydrolysed amino acids above pI

Answer: (A) Both tend net negative.

Why: Above each isoelectric point, deprotonation shifts the balance toward anionic forms. Glycine and alanine side chains do not introduce opposite ordinary ionisation behaviour, and hydrolysis frees their amino and carboxyl groups from the peptide bond.

Why not the others: (B) Positive net charge is favoured at lower pH. (C) The non-ionising side chains do not reverse signs. (D) Net charge changes with pH.

Key point: Evaluate the free products at the stated pH, not the intact peptide.

Q27

Chemistry · medium

Equilibrium / Common-ion solubility approximation

Answer: (C) 1.0×10^{-9} M.

Given: $[Cl^-]\approx0.10$ M from fully dissociated NaCl.

Formula/Principle: $K_{sp}=[Ag^+][Cl^-]$ for AgCl(s).

Step 1: $[Ag^+]=10^{-10}/0.10$.

Step 2: This equals 10^{-9} M, far below added chloride and consistent with the approximation.

Why: Added chloride suppresses silver-ion concentration through the common-ion equilibrium.

Why not the others: (A) Is pure-water solubility; (B) omits division by chloride; (D) reports the chloride concentration.

Key point: Check the approximation after calculation.

Q28

Chemistry · medium

Solutions / Positive deviation from Raoult law

Answer: (B) Weaker unlike interactions.

Why: Mixing replaces some A–A and B–B neighbours with A–B neighbours. When those unlike contacts are less favourable on average, molecules escape more readily than the ideal reference predicts, raising total vapour pressure.

Why not the others: (A) Stronger unlike attractions generally cause negative deviation. (C) Contradicts pressure above a positive ideal prediction. (D) Positive deviation need not mean zero attraction.

Key point: Interpret vapour-pressure deviations through unlike contacts.

Q29

Chemistry · medium

Electrochemistry / Copper sulfate electrolysis with inert electrodes

Answer: (D) Copper and oxygen.

Why: Cu^{2+} gains electrons at the cathode to form solid Cu. The stipulated anodic water oxidation releases O_2; sulfate remains in solution and does not become a gas. The electrode reactions must be assigned separately.

Why not the others: (A) Ignores the specified copper deposition. (B) Sulfate reduction is not the anode reaction. (C) Gaseous sulfate is not an ordinary product.

Key point: Identify reduction and oxidation at their respective electrodes.

Q30

Chemistry · hard

Thermodynamics / Neutralisation calorimetry with limiting reagent

Answer: (A) -100.8 kJ/mol.

Given: Acid 0.0500 mol, base 0.0250 mol, so base limits and water formed is 0.0250 mol.

Formula/Principle: $q_{solution}=mc\Delta T$, $q_{reaction}=-q_{solution}$.

Step 1: $q_{solution}=100\times4.20\times6.0=2520$ J.

Step 2: $\Delta H=-2.520$ kJ/0.0250 mol$=-100.8$ kJ/mol water.

Why: The rising solution temperature means the reaction releases heat; dividing by the limiting amount fixes the molar basis.

Why not the others: (B) divides by all acid; (C) reverses heat sign; (D) halves reacting moles again.

Key point: Determine limiting product moles before normalising calorimetric heat.

Mathematics

Method, reasoning and common errors

Q31

Mathematics · medium

Coordinate Geometry / Tangent-line parameter for an off-origin circle

Answer: (D) 0 and 50.

Given: Completing squares gives centre (3,4) and radius 5.

Formula/Principle: Tangency means centre-to-line distance equals radius.

Step 1: $|3\cdot3+4\cdot4-k|/\sqrt{(3^2+4^2)}=5$, hence $|25-k|=25$.

Step 2: $k=0$ or 50.

Why: The two parallel tangents lie on opposite sides of the circle, yielding both signs of the absolute-value equation.

Why not the others: (A) centres the circle at origin; (B) omits the radius-normal factor; (C) mistakes the centre's line value for a tangent.

Key point: Translate to centre–radius form before using point-to-line distance.

Q32

Mathematics · medium

Differential Calculus / Implicit tangent and intercept

Answer: (C) $x=2$.

Given: Differentiating yields $2x+y+(x+2y)y'=0$.

Formula/Principle: Implicit differentiation gives slope $y'=-(2x+y)/(x+2y)$.

Step 1: At (1,1), slope$=-3/3=-1$.

Step 2: Tangent $y-1=-(x-1)$, or $x+y=2$; set $y=0$ for $x=2$.

Why: The product xy contributes both y and xy'; dropping one term changes the intercept.

Why not the others: (A) sign error in line; (B) reuses point coordinate; (D) ignores nonzero slope.

Key point: Derive slope first, then use the point to form the tangent.

Q33

Mathematics · hard

Differential Calculus / Open-box volume optimisation with a feasible domain

Answer: (B) $(10-2\sqrt{7})/3$ cm.

Given: Height x, base dimensions $12-2x$ and $8-2x$, with feasible $0<x<4$ cm.

Formula/Principle: $V=x(12-2x)(8-2x)=4x(6-x)(4-x)$; maximise inside the physical interval.

Step 1: $V'=4(24-20x+3x^2)$. Setting it to zero gives $x=(10\pm2\sqrt{7})/3$.

Step 2: The plus root exceeds 4 cm and is infeasible; the minus root lies inside (0,4), where V rises then falls. Endpoint volume tends to zero.

Why: A stationary point outside the domain cannot describe a box, even if it solves the algebraic derivative equation.

Why not the others: (A) is the invalid derivative root; (C) is feasible but not stationary; (D) collapses a base dimension to zero.

Key point: Optimisation needs both calculus and feasibility checks.

Q34

Mathematics · medium

Integral Calculus / Difference of logarithmic definite integrals

Answer: (A) $(3/2)\ln2-1$.

Given: The two terms have different denominators and antiderivatives.

Formula/Principle: Substitute $u=1+x^2$ for the first; write $x/(1+x)=1-1/(1+x)$ for the second.

Step 1: First integral $=(1/2)\ln2$.

Step 2: Second integral $=[x-\ln(1+x)]_0^1=1-\ln2$.

Step 3: Difference $=(3/2)\ln2-1$.

Why: On $0<x<1$, $1+x^2<1+x$, so the integrand is positive; the result is approximately +0.040, an independent sign check.

Why not the others: (B) reverses subtraction; (C) omits the second integral; (D) omits the substitution factor.

Key point: Integrate the two rational terms separately, then compare signs.

Q35

Mathematics · easy

Sets, Relations and Functions / Domain of a composition with root and denominator

Answer: (D) $x\geq1$ except 5.

Why: The square root first requires $x-1\geq0$. The outer function additionally forbids its input $t=2$; $\sqrt{(x-1)}=2$ occurs at $x=5$, which must be removed. Both conditions are needed to define the composition.

Why not the others: (A) excludes valid $x=1$ and includes forbidden 5. (B) excludes valid $1\leq x<5$ and includes 5. (C) includes $x<1$ where the root is not real.

Key point: Apply inner-domain and outer-input constraints in order.

Q36

Mathematics · medium

Vector Algebra / Scalar triple product with prescribed volume

Answer: (C) -1 and 5.

Given: Volume is $|a\cdot(b\times c)|$.

Formula/Principle: $b\times c=(-1,t-1,1)$, hence $a\cdot(b\times c)=t-2$.

Step 1: Set $|t-2|=3$.

Step 2: Solve both signs: $t-2=\pm 3$, giving $t=5$ or -1.

Why: Volume loses orientation sign, so two distinct parameter values can create the same magnitude.

Why not the others: (A) Gives zero volume; (B) uses an incorrect centre for the absolute value; (D) misses the negative branch.

Key point: Do not discard either sign of an absolute scalar triple product.

Q37

Mathematics · medium

Matrices and Determinants / Determinant under coupled row combinations

Answer: (B) -6.

Given: Before interchange, the row-combination matrix is T with rows $(1,1,0),(0,1,1),(1,0,1)$.

Formula/Principle: $\det(TA)=\det T\cdot\det A$; one row interchange changes sign.

Step 1: Expand $\det T=1(1)-1(-1)=2$, so intermediate determinant$=2\times 3=6$.

Step 2: Interchange first and third rows to obtain -6.

Why: These are three coupled row combinations, not three independent row additions that each preserve determinant.

Why not the others: (A) omits the final swap; (C) omits the transformation's factor 2; (D) omits both changes.

Key point: Represent simultaneous row combinations by a matrix before applying a later swap.

Q38

Mathematics · medium

Probability / Conditional hypergeometric sample

Answer: (A) 2/11.

Given: Three red draws=C(4,3)=4; exactly two red=C(4,2)C(3,1)=18.

Formula/Principle: Conditional probability is favourable count divided by count satisfying the condition.

Step 1: At least two red includes 18+4=22 selections.

Step 2: All three red probability=4/22=2/11.

Why: After conditioning, the denominator is not all C(7,3)=35 possible samples.

Why not the others: (B) is unconditioned probability; (C) treats two event types as equally likely; (D) uses an incorrect favourable count.

Key point: Build the conditioned sample space before dividing.

Q39

Mathematics · hard

Permutations and Combinations / Lattice paths avoiding a point

Answer: (D) 66.

Given: All paths use five right and four up steps.

Formula/Principle: Count all sequences then subtract those visiting the forbidden point.

Step 1: Total=C(9,4)=126. Paths to (2,2): C(4,2)=6; paths onward: C(5,2)=10.

Step 2: Through-point paths=6×10=60; avoiding=126−60=66.

Why: The two subpaths can combine independently, which calls for multiplication before subtraction.

Why not the others: (A) ignores exclusion; (B) counts forbidden paths; (C) subtracts incorrectly.

Key point: Decompose paths at a required point and use complement for avoidance.

Q40

Mathematics · medium

Binomial Theorem / Vanishing coefficient followed by adjacent term

Answer: (C) −5.

Given: Multiplication by 1−x makes coefficient of x^r equal C(n,r)−C(n,r−1).

Formula/Principle: Set C(n,3)−C(n,2)=0; ratio C(n,3)/C(n,2)=(n−2)/3.

Step 1: (n−2)/3=1 gives n=5.

Step 2: x^4 coefficient=C(5,4)−C(5,3)=5−10=−5.

Why: The vanishing x^3 coefficient fixes the unknown exponent before the requested next coefficient can be evaluated.

Why not the others: (A) reverses subtraction; (B) retains only the shifted term; (D) assumes the next coefficient also vanishes.

Key point: Convolution with a short factor relates adjacent binomial coefficients.

Q41

Mathematics · easy

Sequences and Series / Finite rational telescoping

Answer: (B) 17/30.

Given: Four terms have denominators 3,8,15,24.

Formula/Principle: $1/[k(k+2)]=(1/2)(1/k-1/(k+2))$.

Step 1: Telescoping leaves $(1/2)(1+1/2-1/5-1/6)$.

Step 2: The bracket is 17/15, giving 17/30.

Why: Interior reciprocal terms cancel after shifting by two.

Why not the others: (A) stops at k=2; (C) stops at k=3; (D) accidentally sums $1/[k(k+1)]$ instead of the stated denominator.

Key point: Keep both unmatched terms at each end.

Q42

Mathematics · medium

Quadratic Equations / Root signs plus a threshold

Answer: (A) $0<k<2$.

Given: Root product $k-2$; polynomial value at $x=2$ is $-k$.

Formula/Principle: Opposite root signs require product<0. With an upward-opening parabola and negative root<0, x=2 lies between the roots precisely when $p(2)<0$.

Step 1: $k-2<0$ gives $k<2$.

Step 2: $-k<0$ gives $k>0$, ensuring positive root>2. Combine to $0<k<2$.

Why: At k=0 the positive root equals 2, while k=2 gives a zero root; both strict endpoints are excluded.

Why not the others: (B) includes values whose positive root is below 2; (C) includes forbidden endpoints; (D) loses opposite signs.

Key point: Combine Vieta's sign test with a polynomial-value test at the threshold.

Q43

Mathematics · easy

Statistics / Pooled variance with shifted means

Answer: (D) 8.

Given: Combined mean=(1+5)/2=3.

Formula/Principle: Pooled variance equals average within-group variance plus variance of group means about the combined mean.

Step 1: Within contribution=(4+4)/2=4.

Step 2: Between contribution=$[(1-3)^2+(5-3)^2]/2=4$; total=8.

Why: Groups have different centres, so averaging their individual variances alone omits between-group spread.

Why not the others: (A) omits between-group spread; (B) averages shifts without squaring; (C) double-counts the shift.

Key point: Include both within- and between-group variation.

Q44

Mathematics · medium

Differential Equations / Eliminating constants from a solution family

Answer: (C) $y''=y$.

Why: Differentiating once gives $y'=C_1e^x-C_2e^(-x)$, so a first-order equation $y'=y$ cannot accommodate the second term. Differentiating again restores the plus sign: $y''=C_1e^x+C_2e^(-x)=y$ for all constants.

Why not the others: (A) fails whenever $C_2\neq0$. (B) reverses the second-derivative sign. (D) holds only for the zero function in this family.

Key point: Differentiate enough times to eliminate independent arbitrary constants.

Q45

Mathematics · medium

Limits and Continuity / Continuous extension and derivative at a point

Answer: (B) k=2 and derivative zero.

Given: $\sin(2x)/(2x)\to1$, hence $f(x)\to2$.

Formula/Principle: Continuity fixes k=2; derivative is $\lim[x\to0](\sin2x/x-2)/x$.

Step 1: Using $\sin2x=2x-(2x)^3/6+o(x^3)$, the numerator is $-(4/3)x^2+o(x^2)$.

Step 2: Divide by x to obtain $f'(0)=0$.

Why: A finite continuous value does not itself determine the derivative; a second limiting step is required.

Why not the others: (A) fails continuity; (C) confuses limit with derivative; (D) ignores the standard sine limit.

Key point: Extend continuously before differentiating at a removable point.

Physics

Method, reasoning and common errors

Q46

Physics · easy

Motion in a Plane / Independent vertical motion of projectiles

Answer: (A) Simultaneously.

Why: Both start with zero vertical velocity and accelerate downward at g from the same height. Their vertical displacements are identical functions of time, while horizontal velocity changes only where the first ball lands.

Why not the others: (B) and (C) incorrectly let horizontal motion alter fall time. (D) Horizontal speed affects range but not vertical flight time in this model.

Key point: Resolve projectile motion into independent components.

Q47

Physics · medium

Work, Energy and Power / Instantaneous power as a dot product

Answer: (D) 2 W.

Given: Both force and velocity are vectors with x and y components.

Formula/Principle: $P=F\cdot v$, the rate of work along instantaneous motion.

Step 1: Multiply matching components: $2\times3+(-1)\times4=6-4$.

Step 2: $P=2$ N·m/s$=2$ W.

Why: The force's negative y component removes some of the power added by its x component.

Why not the others: (A) adds absolute contributions; (B) uses speed alone; (C) reverses the dot-product sign.

Key point: Power depends on the component of force along velocity.

Q48

Physics · hard

System of Particles and Rotational Motion / Eccentric cutout moment of inertia

Answer: (C) $13MR^2/32$.

Given: Uniform surface density makes removed mass M/4. The figure gives removed radius R/2 and displacement d=R/2.

Formula/Principle: Subtract the hole's inertia about O using the parallel-axis theorem.

Step 1: Original $I=MR^2/2$; hole's central $I=(M/4)(R/2)^2/2=MR^2/32$.

Step 2: Hole's displacement term $=(M/4)(R/2)^2=MR^2/16$.

Step 3: Remaining $I=MR^2(1/2-1/32-1/16)=13MR^2/32$.

Why: The missing circle is not concentric with O, so subtracting only its central inertia is insufficient.

Why not the others: (A) mishandles removed mass; (B) omits part of parallel-axis shift; (D) treats the hole as central.

Key point: Treat removed material as a negative body at the same axis.

Q49

Physics · medium

Mechanical Properties of Fluids / Free-surface slope in an accelerating tank

Answer: (B) 1/2.

Given: Effective gravity is g down plus g/2 left.

Formula/Principle: A free surface is perpendicular to effective gravity; $|\Delta h/\Delta x|=a/g$.

Step 1: Substitute a=g/2.

Step 2: Slope magnitude $=(g/2)/g=1/2$; it is higher on the left.

Why: In the accelerating frame, the horizontal pseudo acceleration tilts the effective gravity vector.

Why not the others: (A) inverts a/g; (C) assumes a=g; (D) squares the ratio.

Key point: Separate the slope magnitude from its left-right direction.

Q50

Physics · medium

Thermodynamics / Entropy of thermal equilibration

Answer: (A) C ln(4/3).

Given: Energy balance gives final temperature (T+3T)/2=2T.

Formula/Principle: For each body, ΔS=C ln(Tfinal/Tinitial).

Step 1: Add C ln(2T/T)+C ln(2T/3T).

Step 2: Combine to C ln(4/3)>0.

Why: Equal and opposite heat transfer between bodies still creates net entropy when it occurs across a finite temperature difference.

Why not the others: (B) confuses zero net heat with zero entropy; (C) reverses the logarithm; (D) ignores the hot body's entropy loss.

Key point: Sum each body's state-function change.

Q51

Physics · medium

Waves / Spatial phase difference

Answer: (D) π/3.

Given: One full wavelength corresponds to 2π spatial phase.

Formula/Principle: Δφ=2πΔx/λ.

Step 1: Substitute Δx=λ/6.

Step 2: Δφ=2π/6=π/3.

Why: Comparing positions at the same instant isolates the spatial phase term rather than time evolution.

Why not the others: (A) omits the factor two; (B) uses a quarter wavelength; (C) doubles the result.

Key point: Convert displacement to phase as a fraction of a full cycle.

Q52

Physics · medium

Gravitation / Total energy of a circular satellite orbit

Answer: (C) −GMm/(2r).

Given: The circular-orbit condition gives $mv^2/r=GMm/r^2$.

Formula/Principle: $K=mv^2/2=GMm/(2r)$, $U=-GMm/r$.

Step 1: Add K and U.

Step 2: E=GMm/(2r)−GMm/r=−GMm/(2r).

Why: The negative result signifies a bound orbit relative to zero energy at infinity.

Why not the others: (A) loses binding sign; (B) reports U alone; (D) falsely equates kinetic and potential magnitudes.

Key point: Total energy is not the same as gravitational potential energy.

Q53

Physics · hard

Electric Charges and Fields / Zero potential on a signed line

Answer: (B) −a and 3a/5.

Given: The signed-axis diagram fixes charge positions and two sign-change intervals.

Formula/Principle: $V=kq/|x|-4kq/|x-3a|=0$, hence $|x-3a|=4|x|$.

Step 1: For $x<0$, $3a-x=-4x$ gives $x=-a$.

Step 2: For $0<x<3a$, $3a-x=4x$ gives $x=3a/5$; for $x>3a$, the formal solution fails that interval.

Why: Potential is scalar but distances remain positive, so absolute values need separate intervals.

Why not the others: (A) uses equal charge magnitudes; (C) names singular charge positions; (D) chooses distances with incorrect interval signs.

Key point: Verify every algebraic solution in its positional interval.

Q54

Physics · medium

Electrostatic Potential and Capacitance / Plate separation change at fixed battery voltage

Answer: (A) Both become half.

Why: Capacitance $\varepsilon A/d$ halves when d doubles. The connected ideal battery holds V fixed, so $Q=CV$ and $U=CV^2/2$ each scale with C and halve. The battery can accept the charge that leaves the capacitor.

Why not the others: (B) Reverses capacitance change. (C) Assigns different scaling to quantities both proportional to C at fixed V. (D) Treats an attached battery as if it were disconnected.

Key point: Identify the fixed voltage boundary condition before applying capacitance formulas.

Q55

Physics · medium

Current Electricity / Terminal voltage with internal resistance

Answer: (D) 8 V.

Given: Total series resistance 6 Ω.

Formula/Principle: $I=E/(R+r)$, $V_{terminal}=IR=E-Ir$.

Step 1: $I=12/6=2$ A.

Step 2: Terminal voltage $=2\times4=8$ V.

Why: Two volts of internal resistance at 2 A account for 4 V dropped inside the cell.

Why not the others: (A) assumes no internal drop; (B) reports that drop; (C) treats resistance sum as voltage.

Key point: Distinguish emf from voltage under load.

Q56

Physics · medium

Moving Charges and Magnetism / Torque on a current loop

Answer: (C) $(\sqrt{3}/2)IAB$.

Given: Loop normal makes 60° with the field.

Formula/Principle: Torque magnitude $=IAB\sin\theta$, where θ is normal–field angle.

Step 1: $\theta=90°-30°=60°$.

Step 2: $\tau=IAB\sin60°=(\sqrt{3}/2)IAB$.

Why: The magnetic moment is normal to the loop's plane, so the stated plane angle is not directly θ.

Why not the others: (A) substitutes 30°; (B) assumes 90°; (D) assumes alignment of normal and field.

Key point: Convert a plane angle to the dipole-axis angle.

Q57

Physics · medium

Alternating Current / Ideal transformer voltage and power

Answer: (B) 60 V and 2.5 A.

Given: Turns ratio $N_s/N_p=1/4$ and secondary resistor 6 Ω.

Formula/Principle: $V_s/V_p=N_s/N_p$; ideal input power equals output power.

Step 1: $V_s=240/4=60$ V; $I_s=V_s/R=10$ A.

Step 2: $V_pI_p=V_sI_s=600$ W, so $I_p=600/240=2.5$ A.

Why: Lower voltage accompanies higher secondary current for conserved power.

Why not the others: (A) reports secondary current as primary; (C) ignores turns ratio; (D) uses a half instead of quarter ratio.

Key point: Combine turns ratio with load and power conservation.

Q58

Physics · medium

Ray Optics and Optical Instruments / Optical path excess through a slab

Answer: (A) $(n-1)L$.

Given: Slab path length L and refractive index n.

Formula/Principle: Optical path is n times geometric distance in that medium.

Step 1: Slab optical path $=nL$; vacuum path $=L$.

Step 2: Excess $=nL-L=(n-1)L$.

Why: The question asks for a comparison, so total slab optical path alone is not the final answer.

Why not the others: (B) is total path; (C) is not the defined optical path; (D) adds instead of subtracts the reference.

Key point: Subtract the specified baseline from optical path.

Q59

Physics · easy

Electromagnetic Waves / Propagation from E and B orientation

Answer: (D) +x.

Why: Energy transport and propagation follow the Poynting direction E×B. For a right-handed coordinate system, $\hat{y}\times\hat{z}=\hat{x}$, perpendicular to both fields.

Why not the others: (A) Reverses cross-product order. (B) and (C) put propagation along one field, whereas a plane electromagnetic wave is transverse.

Key point: Use field orientation, not field magnitude, for propagation direction.

Q60

Physics · hard

Atoms and Nuclei / Two half-lives: activity and surviving fraction

Answer: (C) 1:2 and 5/32.

Given: X has undergone four half-lives, leaving $N_0/16$; Y two, leaving $N_0/4$.

Formula/Principle: Activity $A=\lambda N$ with $\lambda=\ln 2/T_{½}$; total survivor fraction=$(N_X+N_Y)/(2N_0)$.

Step 1: $\lambda_X/\lambda_Y=2$, so $A_X/A_Y=2(1/16)/(1/4)=1/2$.

Step 2: Surviving fraction=(1/16+1/4)/2=5/32.

Why: More X nuclei have decayed, but its decay constant is twice as large, so population ratio alone is not activity ratio.

Why not the others: (A) uses population ratio as activity; (B) reverses it; (D) reports the decayed fraction.

Key point: Keep population and activity distinct for each isotope.
